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SIMATS ENGINEERING

CO3 - ASSESSMENT TOOL 1

Case Study

COURSE NAME: DEEP LEARNING

COURSE CODE: MLA04

SLOT :

COURSE OUTCOME COVERED

CO3: Analyze and apply convolutional neural network architectures, including convolutional layers, filters, parameter sharing, regularization, AlexNet and ResNet, to develop solutions for image classification and real-world computer vision applications. (L4)

CO Weightage: 25%

Duration: 90 minutes

Assessment Tool Weightage: 25%

Marks: 25

Code of Conduct:

I P.mokshith/192425149(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Name of the student:

Mokshith

P.Mokshith Reddy

Criteria	Conceptual Understanding (25)	Linkages (25)	Organization (2)	Integration of Literature (2)	Clarity & Presentation (1)	Total (10)
Weightage	25 %	25%	20%	20 %	10 %	100%
Q1						
Q2						
Total						

Signature of the Faculty:

Total Marks :

QUESTIONS: 02

Total Marks: 20

Case Study Question 1: CNN for Automated Medical Image Classification (10 Marks)

A healthcare organization wants to develop a deep learning system to automatically classify chest X-ray images into **normal** and **disease-affected** categories. The dataset contains thousands of X-ray images with variations in size, brightness and image quality. A CNN model is proposed because it can automatically learn important visual features such as edges, textures, and abnormal regions without manually designing features.

Question:

1. Design and explain a suitable **CNN architecture** for this medical image classification system.
2. Describe the **motivation for using CNNs** and explain the role of **convolutional layers, filters, pooling layers, and fully connected layers** in the proposed architecture.
3. Explain how **parameter sharing** reduces the number of trainable parameters compared with a fully connected network.
4. Discuss suitable **regularization techniques** such as **dropout, data augmentation and batch normalization** to reduce overfitting.
5. Finally, explain whether **AlexNet or ResNet** would be more appropriate for this application and justify your choice based on feature learning, network depth, computational requirements and classification performance.

Case Study Question 2: CNN-Based Autonomous Vehicle Object Recognition (10 Marks)

An autonomous vehicle company is developing a vision system that must identify **cars, pedestrians, traffic signs, bicycles, and road obstacles** from images captured by cameras mounted on the vehicle. The system must recognize objects under different lighting conditions, viewing angles, distances and road environments. The development team is considering **AlexNet and ResNet** architectures for building the CNN-based recognition system.

Question:

1. Analyze how a **CNN architecture** can be designed for this autonomous vehicle application.
2. Explain the purpose of the **input, convolution, activation, pooling, and fully connected/output layers** and describe how convolutional **filters progressively learn low-level and high-level features**.
3. Explain the importance of **parameter sharing** in reducing computational complexity and improving translation-related feature detection.
4. Discuss the possible problem of **overfitting** and recommend suitable regularization methods.
5. Compare **AlexNet and ResNet** in terms of architecture, depth, parameter efficiency, training difficulty, vanishing-gradient problems and feature-learning capability.
6. Finally, recommend the most suitable architecture for the autonomous vehicle system and justify your decision based on **accuracy, computational cost and real-time application requirements**.

CASE STUDY 1: CNN FOR AUTOMATED MEDICAL IMAGE CLASSIFICATION – 10 MARKS

1. Suitable CNN Architecture – 2 Marks

A suitable CNN architecture is: Input X-ray Image → Convolution → ReLU → Max Pooling → Convolution → ReLU → Max Pooling → Flatten → Fully Connected Layer → Dropout → Output Layer.

The output classifies the X-ray image as Normal or Disease-affected.

2. CNN Layers, Filters and Their Roles – 2 Marks

The input layer receives the X-ray image. Convolutional layers extract important features such as edges, textures and abnormal regions. Filters or kernels detect specific patterns in the image. ReLU introduces non-linearity and helps the network learn complex features. Pooling reduces the spatial dimensions and computational cost. Fully connected layers combine the extracted features, and the output layer predicts the final class.

3. Parameter Sharing – 2 Marks

CNNs use the same filter weights at different locations of an image. A filter that detects an edge can detect that edge anywhere in the X-ray. Therefore, parameter sharing greatly reduces the number of trainable parameters, memory requirements and computational cost compared with a fully connected network.

4. Regularization Techniques – 2 Marks

Data Augmentation: Rotation, scaling, flipping and shifting can increase training data variation.

Dropout: Randomly disables some neurons during training and reduces overfitting.

Batch Normalization: Stabilizes and improves the training process.

Early Stopping: Stops training when validation performance stops improving.

L2 Regularization: Prevents the model from learning excessively large weights.

5. AlexNet vs ResNet – 2 Marks

ResNet is more appropriate for medical X-ray classification. AlexNet is a relatively shallow CNN, while ResNet is deeper and uses residual or skip connections.

ResNet can learn more complex medical image features and reduces the vanishing-gradient problem during deep network training. Although ResNet requires more computational resources, it generally provides better feature extraction and classification performance.

RUBRIC LEVEL: LEVEL 4 – EXEMPLARY

The answer covers CNN architecture, layers, filters, parameter sharing, regularization and AlexNet/ResNet comparison with proper justification.

CASE STUDY 2: CNN-BASED AUTONOMOUS VEHICLE OBJECT RECOGNITION – 10 MARKS

1. CNN Architecture Design – 2 Marks

A suitable architecture is: Camera Image → Preprocessing → Convolution → ReLU → Pooling → Convolution → ReLU → Pooling → Deep Convolution Layers → Detection Layer → Object Classes and Bounding Boxes.

The system can identify cars, pedestrians, traffic signs, bicycles and road obstacles.

2. Purpose of CNN Layers – 2 Marks

The input layer receives camera images or video frames. Convolutional layers extract visual features. Filters detect edges, shapes, textures and object patterns. ReLU introduces non-linearity. Pooling reduces image dimensions while preserving important information. Deeper convolutional layers learn high-level features such as cars, pedestrians and traffic signs. The output or detection layer identifies the object class and its location.

CNNs therefore learn features progressively from low-level features → mid-level features → high-level object features.

3. Parameter Sharing – 2 Marks

Parameter sharing means that the same convolution filter is applied across different locations of an image. For example, a filter detecting a vehicle edge can detect it on the left, center or right side of the image.

This reduces the number of parameters and computational complexity. It also helps the CNN detect objects even when their positions change within the image.

4. Overfitting and Regularization – 2 Marks

Overfitting can occur when the CNN performs well on training images but poorly on new road conditions.

Data Augmentation: Changes brightness, rotation, cropping and weather conditions.

Dropout: Reduces dependence on individual neurons.

Batch Normalization: Improves training stability.

L2 Regularization: Controls large weights.

Early Stopping: Prevents excessive training.

These techniques help the model perform reliably in different lighting, weather and road conditions.

5. AlexNet vs ResNet – 1 Mark

AlexNet is a traditional and comparatively shallower CNN with lower computational requirements. ResNet is deeper and uses residual connections, which make deep network training easier.

ResNet generally provides better feature extraction and accuracy, while AlexNet has lower computational cost.

6. Final Recommendation – 1 Mark

ResNet is recommended for autonomous vehicle object recognition because it provides strong feature extraction and high accuracy.

However, a lightweight or optimized ResNet should be considered because autonomous vehicles require real-time processing, low latency and reasonable computational cost.

RUBRIC LEVEL: LEVEL 4 – EXEMPLARY

The answer demonstrates understanding of CNN architecture, filters, parameter sharing, regularization, AlexNet vs ResNet and real-time autonomous vehicle requirements.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding: clearly identifies all key issues	Good understanding with most issues identified	Basic understanding: some issues missed	Poor understanding: problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity: difficult to understand